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Research on multi-algorithm fusion methods for forecasting typical daily load profiles in power system planning for target years

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Accurately forecasting typical daily load curves is vital for effective power system planning. This study introduces a load curve prediction methodology that integrates multiple algorithms, enhancing both accuracy and reliability. The process begins with the application of the Local Outlier Factor algorithm to cleanse historical data by detecting and replacing anomalies. Next, the Auto-Regressive Integrated Moving Average (ARIMA) model is employed to forecast future daily peak loads, while the Random Forest algorithm predicts the daily normalized load curve, addressing the non-linear characteristics of the load data. Bayesian optimization is utilized to streamline model parameter selection, enhancing both efficiency and predictive accuracy. The daily load curve for the target planning year is derived by combining the peak load values predicted by ARIMA with the normalized curves predicted by Random Forest. Finally, an improved K-means algorithm conducts cluster analysis on the predicted daily load curves for the target year, identifying the mean of the largest cluster as the representative typical daily load curve for each month. Experimental results indicate that this approach effectively captures seasonal variations and trends in load curves, achieving an average prediction error of 0.65% and significantly outperforming individual algorithm predictions.

KEYWORDS

ARIMA, improved k-means algorithm, power system planning, random forest, typical daily load forecasting

1 Introduction

Electric load forecasting forms the foundation of power system planning (Pinheiro et al., 2023). With the continuous development of power systems, the traditional approach of predicting only the peak load is no longer sufficient to meet the future needs of power system planning (Habbak et al., 2023). The rapid growth of wind and solar renewable energy, along with the gradual changes in load characteristics, makes the prediction of daily load curves for the target year and the determination of typical load curves crucial for supporting the development planning of power systems (Wang et al., 2022). From a planning perspective, determining future typical daily load curves helps to more accurately assess

power demand, enhance the reliability of power system planning, inform scientific decision-making for power infrastructure investments, and promote the coordinated development of power sources, grids, loads, and energy storage facilities (Li and Zhang, 2021).

In recent years, research on predicting future typical daily load curves has mainly focused on machine learning and deep learning methods, hybrid prediction models, and clustering analysis. For example, by integrating traditional prediction methods with deep learning load forecasting models, prediction accuracy and stability are improved (Chen and Zhang, 2021). Clustering analysis is used to identify similar load patterns, thereby enhancing the accuracy of power system load forecasting (Yang et al., 2022). Additionally, machine learning techniques are employed to handle data uncertainties and improve load forecasting accuracy (Nasir et al., 2021). Hybrid algorithms have attracted attention for improving load forecasting accuracy by combining the strengths of different models. For instance, a hybrid model combining Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and Random Forest (RF) has been used for forecasting solar power generation (Abumohsen et al., 2024). A deep learning-based Long Short-Term Memory (LSTM) model, combined with a hybrid feature selection method—RMR-HFS-LSTM—reduces overfitting and improves the accuracy of short-term load forecasting (Subbiah and Chinnappan, 2022). The hybrid Convolutional Neural Network - Long Short-Term Memory (CNN-LSTM) model utilizes CNN for feature extraction from high-dimensional data and leverages LSTM to enhance the model's efficiency in time series forecasting for short-term load prediction (Shang et al., 2021; Ullah et al., 2024).

However, these studies have effectively predicted load curves from different perspectives and using various methods, they also have some limitations, such as requiring large amounts of data, high computational resource demands, potential poor performance in handling anomalous data, sensitivity to algorithm selection and parameter settings, susceptibility to noisy data, poor performance in dealing with highly volatile data, and limited model interpretability. Moreover, most current research focuses on short-term load forecasting (Ahmad et al., 2022), with relatively few studies on predicting and determining typical daily load curves for future planning target years, which is the main focus of this paper.

Although significant progress has been made in load forecasting using machine learning and hybrid algorithms, there remains a gap in research that specifically addresses the integration of multiple algorithms to forecast typical daily load curves over long-term planning horizons. Most existing studies focus on short-term forecasting, while long-term, high-dimensional load forecasting with seasonal and cyclic variations has not been sufficiently addressed. This paper fills this gap by proposing a hybrid algorithm that improves forecasting accuracy and reliability for target years in power system planning. As shown in Figure 1, this study proposes a methodology for determining the typical daily load curve for the target year of power system planning, leveraging historical data and integrating multiple algorithms. The process begins with data preprocessing, where missing values are either supplemented or removed, and anomalous data is identified and replaced using the Local Outlier Factor (LOF) algorithm. The daily load data is then normalized. Subsequently, the maximum daily load values from

historical data are extracted, and feature engineering is applied. The Auto-Regressive Integrated Moving Average (ARIMA) algorithm is utilized to forecast the daily peak load for the target year. In parallel, the Random Forest (RF) algorithm predicts the normalized daily load curve for the target year based on normalized historical data, also incorporating feature engineering. The results of the peak load forecast and the normalized load curve prediction are combined to generate the daily load curve for the target year. Finally, an improved K-means algorithm performs monthly clustering, with the mean of the largest cluster representing the typical daily load curve for each month of the target year.

2 Power load data cleaning

During the power load data collection process, issues such as data acquisition equipment failures, transmission device malfunctions, system maintenance, and human errors can lead to missing data or outliers, affecting data quality (Hwang et al., 2022). Therefore, data cleaning is crucial in power load research. Cleaned data can more accurately reflect the actual power load situation, reducing errors in analysis models and enhancing the reliability and stability of predictions. Additionally, data cleaning can effectively improve the training efficiency and performance of prediction models, preventing model biases caused by outliers.

2.1 Missing value processing

In the data cleaning process, the first step is to handle missing values. The mean imputation method is used for this purpose. Given a dataset $X = \{x_1, x_2, \dots, x_n\}$. For missing value x_i , it is replaced with the mean μ of the data points immediately before and after the missing value. If multiple onsecutive data points are missing, the entire load curve is discarded. The mean imputation is shown as Equation 1:

$$\mu = \frac{1}{n} \sum_{j=1}^n x_j \quad (1)$$

Populate the missing values to the mean (Equation 2):

$$x_i = \mu \quad (2)$$

where x_i is the i th data point, μ is the mean of the dataset, and n is the total number of data points in the dataset.

2.2 Outlier handling

Anomalous values are identified using the LOF algorithm. LOF is a density-based outlier detection method. Its basic idea is to identify outliers by comparing the local density of a data point with that of its neighbors. Anomalous values are then replaced using the mean method (Chiu and Fu, 2003). The specific algorithm steps are as follows:

1. Calculate the Euclidean distance between the data points. Let the Euclidean distance between data points P_i and P_j be (P_i, P_j) (Equation 3)

$$d(P_i, P_j) = \sqrt{\sum_{k=1}^m (P_i^k - P_j^k)^2} \quad (3)$$

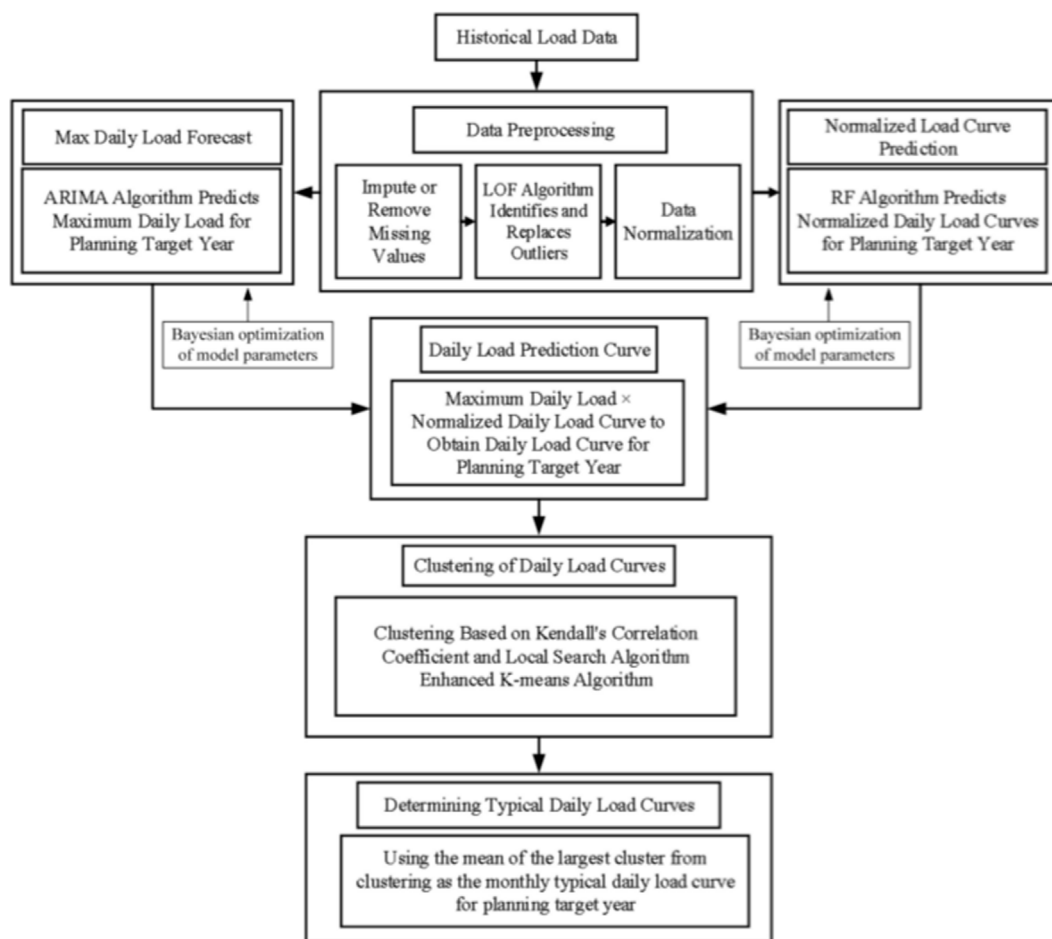


FIGURE 1

Framework for predicting typical daily load curves for the planning target year using multi-algorithm integration.

where: P_i^k is the k th eigenvalue of data point P_i , m is the number of eigenvalues.

- Calculate the k -distance for each data point. For data point P_i , find the distance to its k th nearest neighbor, denoted as $d_k(P_i)$. In this paper, the value of k is set to 20.
- Calculate the Reachability Distance. For data point P_i and its neighbor P_j , the reachability distance is defined as Equation 4:

$$\text{reach_dist}_k(P_i, P_j) = \max(d_k(P_j), d(P_i, P_j)) \quad (4)$$

- For data point P_i , its local reachability density $\text{LRD}_k(P_i)$ is defined as Equation 5:

$$\text{LRD}_k(P_i) = \left(\frac{\sum_{P_j \in N_k(P_i)} \text{reach_dist}_k(P_i, P_j)}{|N_k(P_i)|} \right)^{-1} \quad (5)$$

where $N_k(P_i)$ is the set of k -nearest neighbors of P_i , $|N_k(P_i)|$ is the number of points in the k -nearest neighbors of P_i .

- Calculate the LOF. For data point P_i , its LOF is defined as Equation 6:

$$\text{LOF}_k(P_i) = \frac{\sum_{P_j \in N_k(P_i)} \frac{\text{LRD}_k(P_j)}{\text{LRD}_k(P_i)}}{|N_k(P_i)|} \quad (6)$$

In this paper, a data point is considered an outlier if its LOF value is greater than the threshold of 1.5. Anomalous points are replaced with the mean of the surrounding data points (Equation 7):

$$x_i = \frac{x_{i-1} + x_{i+1}}{2} \quad (7)$$

If $i = 1$ or $i = n$, the outlier is replaced with the adjacent data point.

3 Hybrid algorithm for predicting daily load curves for the target year

3.1 Maximum daily load prediction

The ARIMA (Auto Regressive Integrated Moving Average) model is a widely used statistical model for time series analysis and

forecasting. It combines three components: Auto Regression (AR), Integration (I), and Moving Average (MA), enabling it to capture various patterns in time series data (Zhang, 2003). The ARIMA model effectively captures trends and seasonal variations in time series data, making it well-suited for predicting daily maximum load values (Chen et al., 2022; Chodakowska et al., 2021). However, ARIMA has limitations in handling non-linear relationships, such as the shapes of daily load curves. To improve prediction accuracy, it is beneficial to combine ARIMA with other non-linear models (Zhang et al., 2023; Yu et al., 2021). The method for predicting maximum load based on ARIMA is outlined as follows. Bayesian optimization is a global optimization method commonly used for optimizing objective functions with high computational costs. Unlike traditional optimization methods, Bayesian optimization builds a probabilistic model (typically Gaussian Process) of the objective function and uses this model to select the next experimental points, efficiently exploring the parameter space.

The ARIMA model is expressed as Equation 8:

$$\begin{cases} y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t \end{cases} \quad (8)$$

where p is the number of autoregressive terms, d is the number of differences needed to make the time series stationary, q is the number of moving average terms, y_t is the value of the time series at time t , c is a constant, ϕ_i are the autoregressive coefficients, θ_j are the moving average coefficients, ϵ_t is the error term at time t . In this study, Bayesian optimization is employed to adjust the hyperparameters of the ARIMA model to minimize RMSE. By efficiently navigating the parameter space, Bayesian optimization ensures that the ARIMA model's parameters are optimized for capturing trends and seasonal variations in time series data, thereby enhancing prediction accuracy and model stability.

To obtain the optimal parameters for the ARIMA model, Bayesian optimization is employed, with Root Mean Square Error (RMSE) as the objective function. In this study, Bayesian optimization is employed to adjust the hyperparameters of the ARIMA model to minimize RMSE. By efficiently navigating the parameter space, Bayesian optimization ensures that the ARIMA model's parameters are optimized for capturing trends and seasonal variations in time series data, thereby enhancing prediction accuracy and model stability. RMSE is calculated as follows Equation 9:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (9)$$

Where y_i is the actual value, $\hat{y}_i$ is the predicted value, n is the number of data points.

The objective function is shown as Equation 10:

$$\text{objFunc} = \min_{D, \text{MALags}} RMSE(D, \text{MALags}) \quad (10)$$

In this paper, the parameters are set $D \in [0, 2]$, $\text{MALags} \in [1, 3]$.

3.2 Normalized daily load curve prediction

RF is an ensemble learning method used for classification or regression tasks by constructing multiple decision trees. This

algorithm was proposed by Leo Breiman in 2001 and improves model accuracy and stability by combining the predictions of multiple decision trees (Breiman, 2001).

Daily load curve data often contain complex nonlinear relationships and multidimensional features. Simple linear models may fail to capture these intricate patterns. RF, by integrating multiple decision trees, excels at capturing nonlinear relationships in data and is particularly advantageous in handling high-dimensional data, making it suitable for predicting the shapes of load curves. The daily load data is normalized by dividing each load value by the maximum load value of the day, resulting in a normalized load curve. This step eliminates the impact of the absolute magnitude of daily load values, allowing load curves from different periods to be compared on the same scale. The normalized load curve is more stable, reducing fluctuations caused by changes in absolute load values and enabling the RF model to capture the characteristic shapes of the load curves more effectively. The ensemble learning nature of RF endows it with strong robustness and stability, effectively handling outliers and noise within the load curves (Peng et al., 2021; Fan et al., 2021).

The construction of the RF algorithm involves the following steps:

- Bootstrap Sampling:** Multiple sample subsets are drawn with replacement from the original training dataset. Each subset has the same sample size as the original dataset, but because sampling is done with replacement, each subset may contain duplicate samples.
- Building Decision Trees:** For each sample subset, a decision tree is constructed. During the construction of each decision tree, a random subset of features is selected at each node split (instead of using all features), and the best feature for the split is chosen from this subset. This random feature selection increases the diversity of the model and reduces the correlation between decision trees, thereby enhancing the generalization capability of the ensemble model.
- Ensemble Prediction:** For classification tasks, each decision tree provides a classification result, and the RF determines the final classification result by majority voting. For regression tasks, each decision tree provides a prediction value, and the RF determines the final prediction result by averaging the prediction values from all the trees.

The specific steps are as follows:

Data Normalization. Normalize the daily load curves. Let the load data be $L = \{L_1, L_2, \dots, L_n\}$ and the daily maximum load be $L_{\max}$. The normalization is performed using (Equation 11):

$$L_{\text{normalized}} = \frac{L}{L_{\max}} \quad (11)$$

where L is the daily load data, $L_{\text{normalized}}$ is the normalized load value, and $L_{\max}$ is the daily maximum load value.

- Load Data Prediction:** The RF consists of multiple decision trees. For the RF algorithm, Bayesian optimization is utilized to identify the optimal number of decision trees, the maximum depth of each tree, and other critical hyperparameters. This step ensures that the RF model captures the complex nonlinear relationships and high-dimensional features of the load curves. Bayesian

optimization streamlines the process of hyperparameter selection, improving the overall accuracy and robustness of the RF model. Each tree is constructed using bootstrap sampling from the original dataset, meaning each tree is built from a randomly drawn subset of the data, with replacement. During the construction of each decision tree, only a subset of features is used for splitting nodes, which enhances the model's generalization ability. Suppose there are N decision trees, and each tree's prediction for the load curve is $\hat{y}_i^{(j)}$. The final prediction value is the average of the predictions from where, $\hat{y}_i^{(j)}$ is the prediction value from the j th decision tree for the i th data point, N is the number of decision trees (Equation 12).

$$\hat{y} = \frac{1}{N} \sum_{j=1}^N \hat{y}_i^{(j)} \quad (12)$$

Similarly, Bayesian optimization is used to determine the optimal parameters for the RF model, with RMSE as the objective function. The objective function is shown in Equation 13:

$$\text{objFunc} = \min_{N, D_{\max}, \min_{\text{samples}}} \text{RMSE}(N, D_{\max}, \min_{\text{samples}}) \quad (13)$$

where N is the number of decision trees, $D_{\max}$ is the maximum depth of each tree, $\min_{\text{samples}}$ is the minimum number of samples required to split an internal node.

3.3 Daily load curve prediction

Using a hybrid algorithm of ARIMA and RF to predict the daily load curve for the target year allows us to fully leverage the strengths of both methods.

First, the ARIMA model excels in handling time series data, particularly in capturing trends and periodic features. Through the combination of auto-regressive and moving average components, ARIMA effectively addresses linear and deterministic changes in the time series. Bayesian optimization, using RMSE as the objective function, is applied to optimize the ARIMA model parameters, thereby achieving high-precision predictions of daily maximum load, which lays a solid foundation for constructing the subsequent load curves.

Next, the RF algorithm processes nonlinear relationships and high-dimensional data by integrating decision trees. The daily load curves are normalized, allowing load curves from different days to be compared on the same scale. Bayesian optimization further refines the RF model parameters to ensure accurate predictions of the normalized load curves. RF performs excellently in handling complex nonlinear relationships.

Finally, the complete daily load curve for the target year is obtained by multiplying the daily maximum load predicted by ARIMA with the normalized load curve predicted by RF. The ARIMA model ensures the accuracy of the maximum load, while RF precisely captures the shape of the load curve. This complementary approach results in more reliable and accurate final predictions. The formula for obtaining the complete daily load curve is as follows Equation 14:

$$L_{\text{final}}(t) = L_{\max}(t) \times L_{\text{normalized}}(t) \quad (14)$$

where $L_{\text{final}}(t)$ is the value of the complete daily load curve for day t in the target year, $L_{\max}(t)$ is the predicted daily maximum load value for day t , $L_{\text{normalized}}(t)$ is the predicted normalized load curve value for day t .

4 Selection of typical load profile curves

4.1 Introduction to the K-means algorithm

This paper adopts the mature and straightforward K-means clustering algorithm for analysis. Based on the Euclidean distance, this algorithm effectively clusters curves with similar shapes and sizes, such as daily load profile curves. The K-means algorithm is simple to implement, computationally efficient, and highly scalable (LiKas et al., 2002). However, the algorithm is very sensitive to the selection of cluster centers, and different choices may lead to varying clustering results (Zhao, 2022). Additionally, the K-means algorithm requires the predefined number of clusters to be set in advance, and an inappropriate choice may affect the accuracy of the clustering results (Lei and Li, 2021). While the algorithm iteratively optimizes its solution, it may converge to a local optimum rather than a global one. The improved K-means algorithm used in this study optimizes the traditional K-means method by incorporating Kendall's correlation coefficient and a local search algorithm. This modification reduces the sensitivity of the algorithm to the selection of cluster centers, improving the stability and accuracy of the clustering results. This enhanced method is better suited to handle the complexity of power load data, especially when clustering load curves across multiple months, providing a more reliable foundation for power system planning.

4.2 Improvement of the K-means algorithm

The silhouette coefficient is a clustering evaluation metric used to assess the rationality and quality of clustering results. It compares the similarity within clusters and the dissimilarity between clusters, with values ranging from $[-1, 1]$, where a higher value indicates better clustering effectiveness (Zhou and Gao, 2014) (Equation 15).

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (15)$$

where $a(i)$ denotes the intra-cluster similarity, which is the average distance between the daily load profile sample i and other samples within the same cluster, and $b(i)$ represents the inter-cluster dissimilarity, which is the average distance between sample i and all samples in the nearest neighboring cluster. For the entire daily load dataset, the silhouette coefficient SC can be calculated as the average of all sample silhouette coefficients. Kendall's tau coefficient is a non-parametric method used to measure the degree of correlation between two variables, without assuming any specific distribution for the data (Kendall, 1938). Additionally, it is particularly effective in detecting monotonic relationships, allowing for the identification of consistent increasing or decreasing patterns in load data. This makes it more versatile than methods that assume linearity, such as Pearson correlation. Furthermore, Kendall's Tau provides reliable results even with smaller sample sizes, ensuring that insights can be

drawn from limited data. The Kendall's tau coefficient ranges from $[-1, 1]$, where -1 indicates a perfect negative correlation, 0 indicates no correlation, and 1 indicates a perfect positive correlation. As shown in Equation 16:

$$\tau = \frac{(n_c - n_d)}{\frac{1}{2}n(n-1)} \quad (16)$$

where n_c represents the number of concordant pairs, n_d represents the number of discordant pairs, n is the sample size, and τ is the Kendall's tau coefficient for the load profile curves.

To obtain the daily load profile curves with the least correlation as cluster centers for different k values, solving the minimum correlation problem is similar to solving a shortest distance problem, making it suitable for the use of a local search algorithm. This paper specifically adopts the concept of a Local Search Algorithm to determine cluster centers (Orman et al., 1997). Suppose there are n daily load profile curves, R_s represents the Kendall's correlation coefficient matrix between these curves, and we need to select k curves with the minimum correlation. The specific steps are as follows:

Firstly, randomly select an initial load profile curve y_i and add it to the set of selected curves S .

For the remaining bars, compute the minimum correlation coefficient between them and all the curves in the set S of selected curves (Equation 17).

$$r(j) = \min \{R_s(k, j)\}, k \in S \quad (17)$$

Then, the curve j is randomly selected from the unselected curves, added to the set of selected curves S , and the minimum correlation coefficients between all the remaining curves and all the curves in the set of selected curves S are updated (Equation 18):

$$r(k) = \min \{r(k), R_s(k, j)\}, k \notin S \quad (18)$$

Each time a new curve is selected, evaluate the quality of the current solution and compare it with the previous solution. If the current solution is better, accept it; otherwise, continue searching for other solutions in the neighborhood. By continuously updating the solution, this approach avoids falling into local optima.

Repeat the above process until the number of curves in the selected set S reaches k and meets the specified stopping criteria (such as reaching the maximum number of iterations or the improvement of the solution being less than a threshold). At this point, the sum of the correlations of all curves in the selected set S represents the total correlation among the selected curves.

Repeat the above operations multiple times to select the set of k daily load profile curves with the minimum total correlation as the cluster centers.

Finally, use the selected k daily load profile curves as cluster centers and complete the clustering using the K-means algorithm.

5 Case study analysis

The superiority of the proposed hybrid model over existing singular models (such as standalone neural networks or statistical methods) lies in its "divide-and-conquer" strategy, which

addresses the distinct physical characteristics of power loads. The fundamental rationale is that ARIMA is peak-optimal, while RF is shape-optimal. Specifically, the annual maximum daily load over a long-term planning horizon is heavily influenced by deterministic macroeconomic growth and long-term linear trends. ARIMA, through its autoregressive and differencing mechanisms, is exceptionally well-suited to capture these non-stationary macroscopic trends, yielding optimal peak predictions.

Conversely, the 24-h normalized load shape is driven by complex, high-frequency, non-linear human behaviors and seasonal variations. The Random Forest (RF) algorithm, operating on normalized data, is scale-invariant and immune to the drift of absolute load values. Its ensemble nature prevents overfitting and demonstrates optimal performance in predicting non-linear daily patterns. Conventional models like LSTM or standalone SVR often struggle to simultaneously balance the absolute magnitude shift over years and the highly volatile daily shape. Our fusion approach decouples these two dimensions, mathematically guaranteeing that neither the trend nor the specific contour is compromised, thereby outperforming individual models in long-term typical curve forecasting.

The case study is based on the MATLAB 2022b platform, using the actual daily 24-h load data from the Western Inner Mongolia Power Grid in China from 2019 to 2023 as the basis. The load data is extracted from the aggregated data of all smart meters in the grid and exported through the D5000 dispatch control system of the grid dispatching department. The electricity load in this region is mainly industrial, accounting for 80%. Over the past 5 years, the average annual load growth rate has reached 10%, with typical dual peaks during midday and nighttime, and a daily peak-to-valley difference of less than 8%. The method for determining typical daily load curves for the year 2028, using a multi-algorithm fusion approach for power system planning objectives, predicts monthly typical daily load curves with a data collection interval of 1 h.

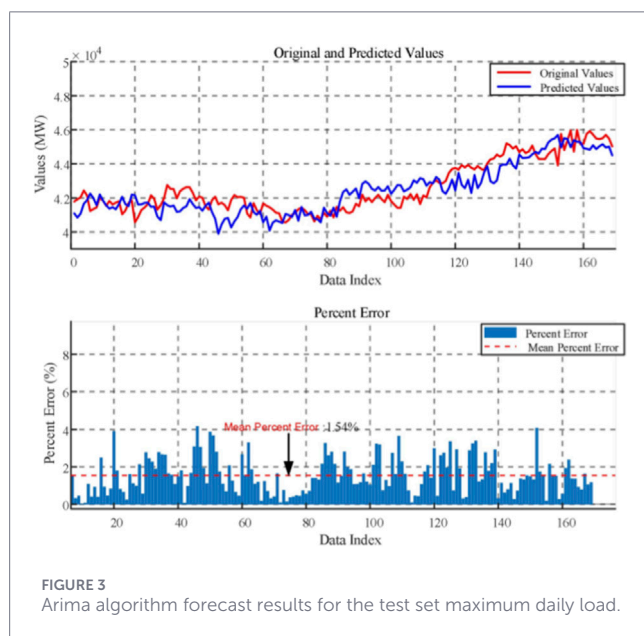
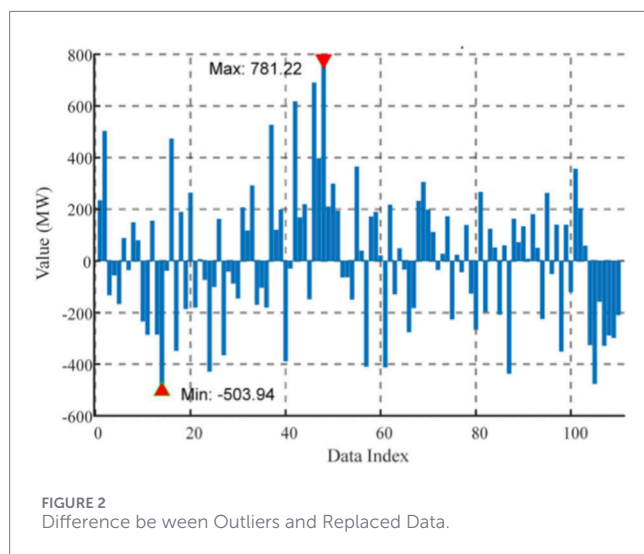
5.1 Data cleansing

Using the methods described in Section 2 for handling missing values, detecting and replacing outliers, the original data is cleansed. Firstly, missing values are replaced using (Equations 1, 2). Subsequently, outliers are detected using the LOF algorithm, and then (Equation 7) is applied to replace the detected outlier data. Figure 2 illustrates the differences between the algorithm-detected outlier data and their replacement data.

A total of 110 outliers were detected, with differences between replaced data and outlier values ranging from 781.22 MW to 503.94 MW. This illustrates that through data cleansing, data quality can be significantly enhanced, improving the accuracy and reliability of the data. This provides a more reliable foundation for subsequent load data analysis and forecasting.

5.2 Maximum load forecasting

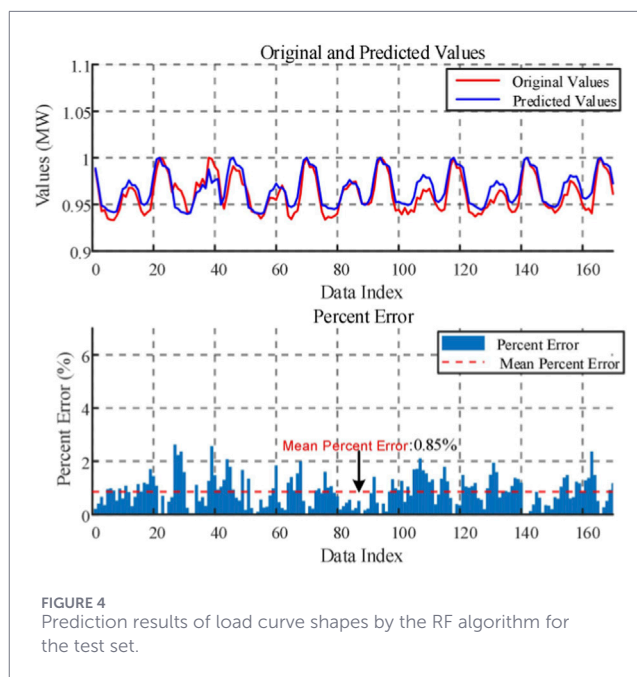
A total of 1826 daily maximum load data points were extracted from 2019 to 2023. Among these, 80% were used for training and 20% for testing. Utilizing the ARIMA algorithm discussed in Section 3, this paper forecasts the annual maximum load for the target year 2028 based on a 5-year period. The last 168



data points of the test set, consisting of both original values and predicted values, are shown in Figure 3. The overall trend of the predicted maximum load aligns with the original load, with an average prediction deviation of 1.54%. From a planning perspective, this 1.54% average prediction deviation indicates high accuracy, suggesting that the predicted results closely match the actual load. Although there are fluctuations at individual points, there is no systematic bias, indicating good stability of the model's predictions at different time points. This capability provides reliable estimates of electricity demand, assisting in the formulation of rational power system planning and resource allocation strategies.

5.3 Daily load curve shape prediction

Firstly, Equation 11 is used to normalize historical daily load curves to extract their shapes. There is a total of 1826 normalized

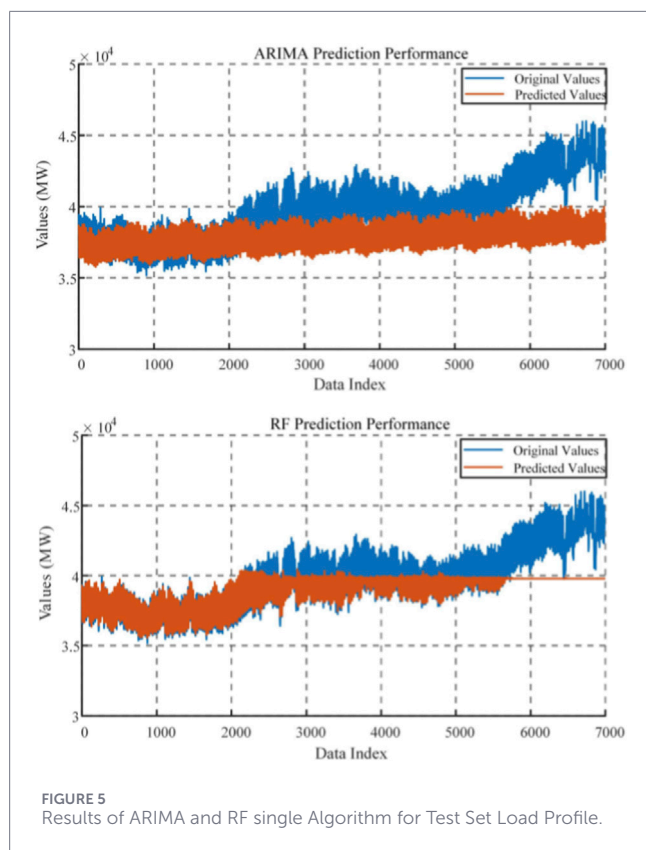


daily load curves, with 80% used for training and 20% for testing purposes. Using the RF algorithm discussed in Section 3, the normalized daily load curves are used to predict the daily normalized load curves for the year 2028. Bayesian optimization is employed to determine the parameters of the RF algorithm. The last 168 data points of the test set, representing 1 week of actual and predicted values, are shown in Figure 4. The predicted normalized load curve shapes closely match the original normalized load curve shapes, with an average prediction error of 0.85%. This indicates a high overall prediction accuracy, meeting the requirements for power system planning research.

5.4 Daily load curve prediction

To demonstrate the effectiveness of the proposed algorithms, predictions were directly made using the ARIMA and RF algorithms. 80% of the historical data was used as the training set, and 20% as the test set. The original values and predicted values of the test set are shown in Figure 5 for comparison.

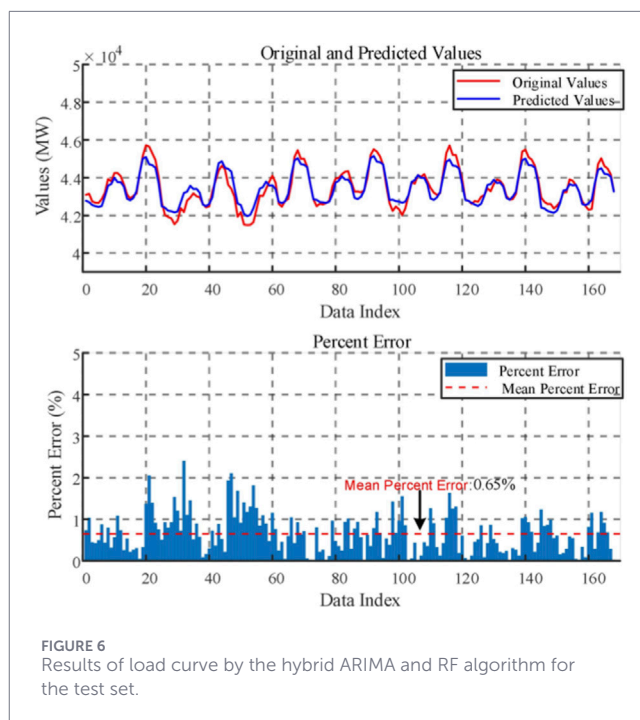
It can be seen that direct predictions using the ARIMA and RF algorithms are not satisfactory. The computational complexity of the ARIMA model increases with the volume of data and the length of seasonal cycles. For hourly data, there are 8760 data points per year, and the seasonal cycle also consists of 8760 data points, resulting in significantly increased model training and prediction times. Although hourly load data also exhibit annual periodicity, their greater volatility makes them more challenging to capture. With large data volumes and long cycles, the applicability of the ARIMA model decreases, leading to longer computation times and potentially poor prediction results. RF is a powerful non-parametric model capable of capturing complex nonlinear relationships. RF performs well in predicting datasets with stable data patterns and trends. However, in long-term forecasting, RF may perform poorly because it lacks inherent time



series processing mechanisms such as autoregressive properties or memory effects. When predicting beyond the training data range, RF may degrade to predicting the mean or average trend of the training set. It can be observed that direct predictions using ARIMA and RF algorithms yield unsatisfactory results for load data with large fluctuations and challenging to capture periodicity and trends.

In scenarios with smaller data volumes and clear data cycles, the ARIMA model can effectively train and predict, making it suitable for predicting maximum daily load data. For predicting load curve shapes, RF can effectively learn the relationships between date, hour features, and load shapes. Normalization allows load curves for each day to vary within the same range, reducing the variability in load changes between different days and making the shapes easier to predict. Predicting normalized load curve shapes using the RF algorithm is superior to directly predicting time series loads. Combining the maximum load predicted by the ARIMA model with the normalized load curve predicted by RF using [Formula 14](#), the last 168 data points of the test set, representing 1 week of actual and predicted values, are shown in [Figure 6](#).

To further illustrate the effectiveness of the proposed method, predictions were made using four models: ANN (Artificial Neural Network), DTR (Decision Tree Regression), GBR (Gradient Boosting Regression), and SVR (Support Vector Regression). The RMSE, MAE, and MAPE of the actual and predicted values for the last 168 data points were compared. The model parameters and prediction results are shown in [Table 1](#).



5.5 Baseline models overview for comparison

To rigorously validate the performance of the proposed hybrid algorithm, several mainstream machine learning and statistical models are introduced as baselines: Artificial Neural Network (ANN), Decision Tree Regression (DTR), Gradient Boosting Regression (GBR), and Support Vector Regression (SVR). ANN models utilize interconnected layers of neurons to approximate complex non-linear functions, though they can be prone to overfitting in noisy load data. SVR maps input features into high-dimensional spaces using kernel functions to find the optimal regression hyperplane, which is robust but highly sensitive to hyperparameter selection. Tree-based models like DTR and GBR handle non-linearities by partitioning the feature space; however, single decision trees lack stability, and gradient boosting approaches can be computationally intensive for high-frequency hourly data.

In addition to these, deep learning models such as Long Short-Term Memory (LSTM) networks are widely cited in literature for time-series forecasting. While LSTM excels at capturing short-term sequential dependencies, it often suffers from memory decay and error accumulation when applied to long-term planning targets (e.g., forecasting years into the future). By comparing our method against these established baselines, the stability and accuracy advantages of our proposed ARIMA-RF fusion framework become clearly evident.

5.6 Determination of typical daily load curves

The proposed combination of ARIMA and RF algorithms is used to predict the daily load curves for the target year 2028, as shown in [Figure 7](#).

TABLE 1 Prediction results from multiple algorithms.

Algorithm	Main parameters	RMSE	MAE	MAPE
ANN (Artificial Neural Network)	hiddenLayerSize:30 trainFcn: Trainlm performanceFcn: mse max_fail:6	733.56	606.67	1.41%
DTR (decision Tree Regression)	MaxNumSplits:20 MinLeafSize:1 MaxDepth: Inf Surrogate: off	2941.35	2708.83	6.19%
GBR (gradient Boosting Regression)	Method: LSBoost NumLearningCycles:100 Learners: tree Learn Rate:0.1 MaxNumSplits:20	3821.42	3642.72	8.35%
SVR (support Vector Regression)	Kernel function: Gaussian Standardize: TRUE Cost:1	1156.53	11509.92	26.52%
Hybrid RF and ARIMA	--	351.27	283.13	0.65%

It can be seen that the proposed method outperforms the predictions of ANN, SVR, DTR, and GBR., Although ANN, also delivers relatively good prediction results, its RMSE, MAE, and MAPE, values are still higher than those obtained using the Hybrid RF, and ARIMA, method proposed in this paper.
Bold values indicate the best performance for each metric.

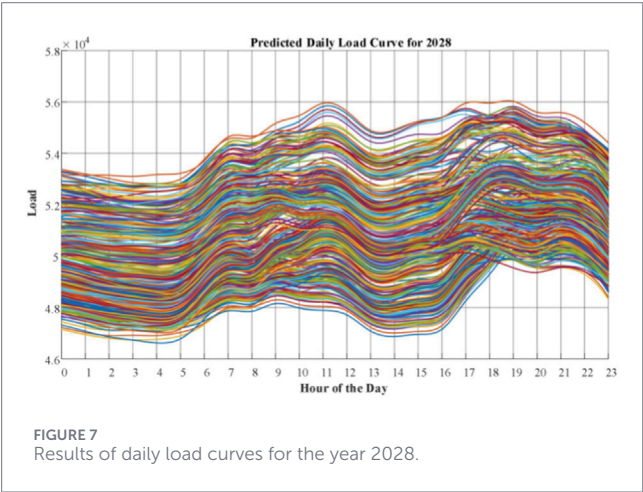


FIGURE 7 Results of daily load curves for the year 2028.

Daily load curves for December 2028 are selected for cluster analysis to demonstrate the effectiveness of the improved K-means algorithm proposed in this paper. Firstly, direct clustering is performed using the K-means algorithm, with k values ranging from two to 6. Each k value undergoes five clustering iterations, and the average silhouette coefficient for each clustering is calculated. The clustering results are presented in Table 2.

TABLE 2 Direct clustering results using k-means algorithm.

k	SC1	SC2	SC3	SC4	SC5
2	0.64	0.64	0.67	0.64	0.67
3	0.53	0.46	0.41	0.46	0.70
4	0.23	0.42	0.39	0.41	0.42
5	0.35	0.38	0.44	0.47	0.43
6	0.43	0.46	0.25	0.25	0.46

TABLE 3 Clustering results using k-means algorithm.

k	2	3	4	5	6
SC	0.67	0.7	0.53	0.51	0.52

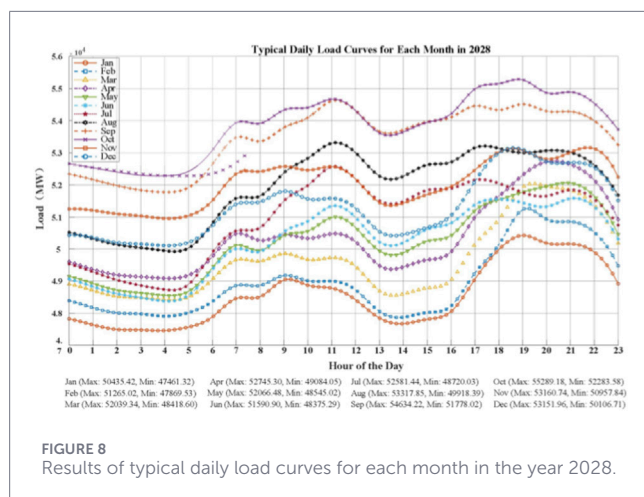
As shown in Table 2, direct clustering using K-means results in unstable clustering outcomes due to the random selection of cluster centers.

Improved K-means clustering is performed by first calculating the Kendall's tau coefficient between curves. A local search algorithm is used to find k curves with the minimum correlation for different k values, which are then selected as cluster centers. The clustering results are presented in Table 3.

Furthermore, to deeply validate the effectiveness of the improved K-means algorithm, we compared its performance against two other prominent clustering algorithms widely used in the literature: Density-Based Spatial Clustering of Applications with Noise (DBSCAN) and Hierarchical Clustering. Using the forecasted 2028 daily load profiles as the input dataset, we calculated the Silhouette Coefficient (SC) for each method. DBSCAN struggled with the uniform density of the normalized curves, yielding an average SC of 0.38. Hierarchical Clustering performed better, achieving an SC of 0.45 at optimal tree cuts. In contrast, our improved K-means algorithm, enhanced by Kendall's Tau correlation and local search initialization, maintained an SC above 0.51 across various k values (as shown in Table 3). This comparative analysis demonstrates that our improved approach significantly mitigates cluster overlap and provides the most distinct and representative typical load profiles.

It can be seen that the improved K-means algorithm proposed in this paper enhances clustering results significantly by optimizing the method of selecting cluster centers, improving robustness. By clustering daily load curves into typical monthly load profiles, the improved K-means method helps power system planning reflect the load characteristics of different seasons and months more accurately. This provides better load forecasting tools for power companies and grid operators, helping to optimize energy generation and distribution. Particularly with the increase in renewable energy sources, the improved K-means algorithm offers significant advantages in identifying and addressing the variability and uncertainty in load curves, further enhancing the reliability and efficiency of power systems.

Using the improved K-means algorithm proposed in this paper, the daily load curves predicted for 2028 are clustered by



month. The average of the largest cluster's results is taken as the monthly typical daily load curve. The typical daily load curves are shown in Figure 8.

It is evident that the load curves for different months in this region exhibit distinct patterns. Higher loads in winter and summer correspond to increased heating and cooling demands, respectively. The lower peak-to-trough ratios of each typical day reflect the region's characteristic of a high proportion of stable industrial loads, aligning with actual electricity usage patterns. The curves obtained through a combination of ARIMA and RF models ensure accuracy in both trends and shapes, providing a solid data foundation for power system planning.

5.7 Robustness and sensitivity analysis

Since the proposed methodology is validated primarily on the Western Inner Mongolia Power Grid dataset, it is imperative to verify the model's robustness against data uncertainty and its sensitivity to hyperparameter variations.

First, to test robustness, Gaussian white noise with varying Signal-to-Noise Ratios (SNR of 5% and 10%) was artificially injected into the historical load data. The model was retrained on this noisy dataset. Experimental results indicate that the Mean Absolute Percentage Error (MAPE) of the hybrid model remained stable, increasing only marginally from 0.65% to 0.71% (under 5% noise) and 0.82% (under 10% noise). This confirms that the initial LOF data cleansing and the ensemble nature of the RF algorithm effectively buffer against data volatility.

Second, a sensitivity analysis was conducted on the core hyperparameter of the RF model: the number of decision trees (N). We varied N from 50 to 300. The objective function (RMSE) exhibited significant fluctuations when $N < 100$, but stabilized to a near-constant minimum once N exceeded 150. This demonstrates that the algorithm is not overly sensitive to precise parameter tuning within the optimal range determined by Bayesian optimization, ensuring high reliability when generalizing to other regional datasets.

6 Discussion and conclusions

This study presents a multi-algorithm fusion framework designed to forecast typical daily load profiles for long-term power system planning. The experimental results validate the effectiveness of the proposed "divide-and-conquer" strategy, which decouples the forecasting of load trends from load shapes. Direct application of single algorithms proved insufficient for this complex task: the ARIMA model, while effective for trends, faces high computational costs with hourly data, whereas the Random Forest (RF) model lacks the memory mechanism required for long-term extrapolation. By integrating ARIMA for daily peak loads and RF for normalized curve shapes, the hybrid model achieved a prediction error (MAPE) of only 0.65%, significantly outperforming ANN (1.41%), DTR (6.19%), and SVR (26.52%). This demonstrates that integrating linear trend capturing with non-linear shape fitting is a superior strategy for target-year planning compared to individual models.

Furthermore, the stability of the clustering method is critical for determining reliable planning baselines. The standard K-means algorithm exhibited high instability in our analysis, with Silhouette Coefficients fluctuating between 0.23 and 0.67 due to random initialization. In contrast, the improved K-means algorithm—incorporating Kendall's Tau correlation and a local search algorithm—stabilized the coefficients above 0.51. This robustness ensures that the selected typical load curves are reproducible and truly representative of seasonal characteristics, mitigating the risk of inconsistent planning decisions caused by algorithmic randomness.

In conclusion, this paper fills a research gap in long-term typical daily load curve forecasting. The case study on the Western Inner Mongolia Power Grid, characterized by an 80% industrial load share, confirms the method's practical applicability. The generated curves accurately reflect distinct seasonal patterns and peak-to-valley characteristics, providing a solid data foundation for infrastructure investment and renewable energy integration.

Future research may explore the application of this hybrid methodology across different regions and load types to validate its versatility and effectiveness further.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Author contributions

QG: Conceptualization, Writing – original draft, Methodology, Project administration. YeZ: Validation, Formal Analysis, Conceptualization, Writing – original draft. QH: Data curation, Writing – review and editing, Investigation. CW: Project administration, Validation, Resources, Writing – review and

editing. YiZ: Writing – review and editing, Conceptualization, Investigation, Data curation, Formal Analysis, Methodology. SH: Supervision, Writing – review and editing, Methodology, Formal Analysis, Data curation. KH: Investigation, Conceptualization, Funding acquisition, Formal Analysis, Writing – review and editing, Supervision.

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Conflict of interest

Authors YeZ, QH, CW, and YiZ were employed by Inner Mongolia Power Electric Operation Control Company.

The remaining author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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